

DSA210 Project Proposal

Can Google Trends Help Predict Short-Term Bitcoin Direction?

Title

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Project Overview

This project examines whether online public attention, measured through Google Trends, can improve short-term prediction of Bitcoin price direction. The study combines historical Bitcoin market data with Google Trends search-interest data, applies preprocessing and feature engineering, and evaluates whether attention-based variables add predictive value beyond standard market indicators.

Research Question

Can Google Trends data meaningfully improve next-day Bitcoin up/down prediction compared with market-only models?

Motivation

Bitcoin is one of the most visible and volatile financial assets, and its price behavior is often influenced by public attention as well as market fundamentals. Search activity may serve as a proxy for investor interest, curiosity, or concern. This project aims to test whether shifts in public attention contain useful information for short-term market prediction.

Understanding this relationship may help clarify whether search trends are relevant primarily for directional forecasting or whether they are more strongly associated with periods of higher market volatility.

Data Sources

1. CoinGecko

Used to obtain:

- Bitcoin price
- Market capitalization
- Total trading volume

2. Google Trends

Used to obtain:

- Search interest for the keyword Bitcoin
- A proxy measure of public attention

Methodology

The project follows the pipeline below:

1. Market data preparation

Clean the raw CoinGecko export and construct a daily Bitcoin market dataset.

2. Google Trends integration

Merge weekly Google Trends data with the cleaned daily Bitcoin market data.

3. Trend-value filling

Restrict the analysis period to dates with available Google Trends data and forward-fill weekly values across daily observations.

4. Feature engineering

Create variables such as:

- daily return
- 7-day rolling volatility
- price change
- Google Trends change
- next-day binary target (up/down)

5. Exploratory data analysis and hypothesis testing

Examine relationships between Google Trends and market outcomes using correlation analysis and group comparisons.

6. Machine learning modeling

Train and compare two classification models:

- Logistic Regression
- Random Forest

7. Performance comparison

Evaluate whether adding Google Trends features improves predictive accuracy relative to market-only models.

Preliminary Findings

The current results suggest that Google Trends variables do **not** meaningfully improve next-day Bitcoin direction prediction.

Statistical findings

- Correlation between Google Trends score and daily return: -0.0813
- p-value: 0.0005
- Correlation between Google Trends score and 7-day volatility: 0.4966
- p-value: < 0.001
- Difference in daily return between high-trend and low-trend days: not statistically significant
 - t-statistic: **-0.5404**

- p-value: **0.5890**

Machine learning findings

- Logistic Regression (market only): 0.5096 accuracy
- Logistic Regression (with Google Trends): 0.5096 accuracy
- Random Forest (market only): 0.4904 accuracy
- Random Forest (with Google Trends): 0.4849 accuracy

Interpretation

The preliminary evidence suggests that Google Trends is weakly related to daily returns but moderately associated with market volatility. Adding Google Trends variables did not improve predictive accuracy in the current models, and in the Random Forest model performance decreased slightly.

These results indicate that search activity may be more informative about market attention, excitement, or turbulence than about short-term directional price movement.

Repository Structure

The repository is organized as follows:

- ``data/raw/`` - original CoinGecko and Google Trends files
- ``data/processed/`` - cleaned, merged, and feature-engineered datasets
- ``data/outputs/`` - summary tables and figures
- ``src/`` - preprocessing and results-generation scripts
- ``notebooks/`` - exploratory analysis, hypothesis testing, machine learning, and plotting scripts
- ``README.md`` - project overview and reproducibility instructions
- ``requirements.txt`` - package dependencies

Limitations

- Google Trends captures public attention rather than true market sentiment.
- The project currently relies on Google Trends only; news and social-media sentiment are not included in the final models.
- Model evaluation is based on a single train/test split.
- More advanced time-series validation could improve robustness.

Planned Improvements

- Add GDELT-based news features
- Add lagged Google Trends variables
- Use time-series cross-validation
- Test additional models and evaluation metrics
- Prepare a summary notebook for clearer presentation of results

Expected Contribution

This project contributes to the question of whether publicly observable attention signals can improve short-term cryptocurrency forecasting. Even if Google Trends does not enhance next-day

direction prediction, the findings are still valuable in showing that attention may be more useful for understanding volatility than for forecasting returns.

Model Summary Table

Model	Feature Set	Accuracy
Logistic Regression	Market only	0.5096
Logistic Regression	With Google Trends	0.5096
Random Forest	Market only	0.4904
Random Forest	With Google Trends	0.4849